Title

SUBJECT

Time Step t:

Observation $o_t \in \mathcal{O}$

Action $a_t \in \mathcal{A}$ using a policy $\pi(a_t | c_t)$, while $c_t = (o_1, a_1, \cdots, o_{t-1}, a_{t-1}, o_t)$

(context)

Problem: Learning a policy π is difficult when the mapping $c_t \mapsto a_t$ is highly implicit. (t is big)

ReAct: $\hat{\mathcal{A}} = \mathcal{A} \cup \mathcal{L}$, $\mathcal{L}$ is the space of language

$\hat{a}_t \in \mathcal{L}$ is a <thought> which doesn't affect the environment

$\xrightarrow{\text{update}} c_{t+1} = (c_t, \hat{a}_t)$